

Anusha Ravindra

anusharavindra21@tamu.edu | www.linkedin.com/in/anusharavindra21 | +1(979) 574 9948

PROFESSIONAL SUMMARY

Electrical engineer with expertise in electric machine design, FEA modeling, and simulation automation. Developed rare-earth-free motor technologies achieving significant material cost savings and reduced design cycle time substantially.

EDUCATION

- Texas A&M University, College Station, TX** United States
Master of Science in Electrical Engineering; GPA: 4.0 Dec 2026
Coursework: Power Electronics, DSP-Based Electromechanical Motion Control, Electronic Motor Drives, Linear Multivariable Systems, Computer Aided Design of Electromechanical Motion Devices
- Sapthagiri College Of Engineering, Karnataka** India
Bachelor of Technology in Electrical and Electronics Engineering; GPA: 3.628 June 2023

SKILLS

- Programming Languages:** Python, MATLAB, PyAnsys, C++.
- Software:** AutoCAD, SIMULINK, Ansys Maxwell 2D, Ansys Maxwell 3D, Ansys MotorCad, Solidworks, Simcenter MotorSolve, Simcenter Magnet, PLECS, Altium, ElmerFEM, Gmsh, MultiSim, Microsoft Excel, PowerPoint, Word.
- Analytical Concepts & Coursework:** Electromagnetic Simulation and Optimization, Finite Element Analysis (FEA), Electric Motor Design, Automation of Simulation Workflows, Technical Documentation, Performance Verification against Simulated Data, 3D Printing.

EXPERIENCE

- Toshiba International Corporation** Houston, TX, United States
R&D Electrical Engineering Intern May 2026 – Present
 - Improving customer test report delivery process for industrial induction motors by converting macro-based files into macro-free formats, enabling smoother workflow, easier review, and reduced process complexity.
 - Collaborating with test and electrical engineering teams to understand Method F induction motor testing procedures and align customer deliverables with engineering requirements.
 - Working with the production team to review and update nameplate information for industrial induction motors, ensuring accuracy in motor specifications and production documentation.
 - Supporting R&D analysis of transient torque behavior in industrial induction motors under fault conditions to evaluate motor performance during dynamic operating states.
- Advanced Electrical Machine and Power Electronics Lab, Texas A&M University** Texas, United States
Student Graduate Worker September 2025 – Present
 - Supporting research on the design and electromagnetic modeling of an Asynchronously Excited Brushless Wound-Field Synchronous Machine.
 - Performing FEA-based electromagnetic analysis, performance evaluation, and design optimization.
 - Developing code to interpolate datasets and extract parameters from lookup tables for motor analysis and control.
 - Working on the logic and code to analyse data and plot Torque-speed Curve and Efficiency Map of an Asynchronously Excited Brushless Wound-Field Synchronous Machine.
- Chara Technologies Pvt. Ltd** Karnataka, India
Electromagnetic Design Engineer June 2023 – June 2025
 - Designed rare-earth-free motors achieving 40% cost reduction while maintaining efficiency.
 - Automated motor simulation workflows using Python and MATLAB for Ansys Maxwell and Simcenter Magnet.
 - Implemented open-source simulation pipelines using Gmsh and ElmerFEM, reducing licensing costs by 90%.
 - Developed multi-objective optimization algorithms including Gradient Descent and Genetic Algorithms.
 - Designed, prototyped, and tested a 5HP ferrite-assisted SynRM achieving IE5 efficiency.
 - Applied field-oriented control (FOC) concepts for synchronous reluctance motors.

ACADEMIC PROJECTS

- Three-Phase BLDC Motor Drive Implementation (Aug–Sep 2025):**
 - Implemented trapezoidal commutation on a TI Delfino C2000 MCU with Hall sensor feedback and ePWM inverter control.
 - Developed PI speed control and fixed-distance speed estimation using external interrupts.
 - Conducted hardware–software co-design, oscilloscope-based transient analysis, and motor pole identification.

PATENT

Filed a patent application for “**Novel motor rotor design with peripherally disconnected flux carriers of Ferrite Assisted Synchronous Reluctance Motors to enhance efficiency and minimize rotor iron loss**”. Currently under review.